CS699, Spring 2026

Project Assignment

The project is a practice of building and testing classification models using a real-world dataset. The goal of the project is to let students explore different methods to build high-performance classification models. The project is performed by a team of two students.

You must read this entire assignment carefully before you start the project.

Dataset

The original data was downloaded from the 2023 American Housing Survey (https://www.census.gov/programs-surveys/ahs/data.html?text-list-0ff840ba34%3Atab=2023#text-list-0ff840ba34) and it was modified for this assignment.

The given dataset *project_data.csv* has 9241 tuples and 145 variables.

In the dataset, each tuple is a household and *Class* is the class attribute. The class attribute value Yes means the household has at least one person with disability (hearing disability, visual disability, etc.). Meanings of other variables can be found in the codebook, which is posted on Blackboard.

Goal

The goal of the project is, using the given dataset, to build multiple classification models, compare their performances, and select the "best" model.

Grade Breakdown

- Intermediate report: 10%
- Presentation: 10%
- Project overall (model performance, project report quality, and R code): 80%

Timeline

- Intermediate report due: 2/25
- Final report due: 4/8
- Presentation slides due: 4/15
- Presentation: 4/29

Intermediate Report Requirement and Grading

- You need to study the dataset carefully and you need to consult the codebook to understand the meanings and values of variables.
- You must perform data exploration and all preprocessing.
- Include in your report a detailed description of all data exploration and preprocessing you performed.
- You must also submit your R or Python code.

- Combine the intermediate report file and the R or Python code in a single archive file and name it *LastName_FirstName.ext*, where *ext* is an appropriate file extension, such as *zip* or *rar*. Include names of both team members. Submit the archive file.
- If not sufficient data exploration and preprocessing is performed, up to 6 points will be deducted.
- If your report is poorly written, not detailed enough, or not well organized, up to 4 points will be deducted.

<u>Project Process Requirement and Grading</u>

- You may use R or Python.
- In general, you may use any preprocessing method and you may use any classification algorithm to find a model with high performance.
- You must use at least four methods to create balanced datasets.
- Optionally, you may apply feature selection methods on your training dataset to reduce the number of variables.
- You must use at least nine different classification algorithms.
- Refer to a simplified overall process diagram attached at the end of this assignment.
- When you write your code, you must follow the requirement in the "Project code submission requirement" section at the end of this assignment.
- You must perform parameter tuning when trying to find high performance models.
- You must handle class imbalance issues using only balanced training datasets. You must not use any other methods:
    - You must use 0.5 as the classification threshold. You must not change the threshold (e.g., during parameter tuning)
    - You must not assign different weights to different classes when estimating models' performance
    - You must not use a cost-sensitive method to handle class imbalance (by assigning different costs to different types of classification errors)
    - Any other methods (except data balancing methods)
- You must do your best to achieve high performance, such as:
    - try different methods to create balanced training datasets
    - try different classification algorithms
    - try parameter tuning
    - etc.
- If there is no clear evidence showing that a team did their best to achieve high performance, up to 20 points will be deducted.
- If your performance results are not reliable, at least 20 points will be deducted.

Project Performance Requirement and Grading

- Performance of your best model (not all models) is considered.
- Minimum requirement
  - class Yes TPR >= 71% AND class No TPR >= 67%
- If the minimum requirement is not met, 10% will be deducted.
- If performance satisfies the following criterion, extra credit of 10% will be given.
  - class Yes TPR >= 74% AND class No TPR >= 70%
- If performance satisfies the following criterion, extra credit of 20% will be given.
  - class Yes TPR >= 77% AND class No TPR >= 73%
- If I cannot reproduce performance measures of your models:
  - At least 20% will be deducted.
  - If the team received any extra credit, the extra credit is revoked (in addition to 20% deduction)

Project Report Requirement and Grading

- Your final report must include:

  - Cover page (including the names of team members)
  - Brief description of data mining tool(s) you used.
  - Brief description of all classification algorithms you used.
  - ***Detailed*** description of data mining procedure (the procedure you actually followed) including all data preprocessing you performed. This must be a step-by-step description which I may have to rely on to reproduce your results.
  - Data mining result and evaluation:
    - Performance measures of all models you built.
      For each model, you must include the following performance measures:
      - **Confusion matrix**
      - **Performance measure table:** Show, for each class, TP rate, FP rate, precision, recall, F-measure, ROC area, MCC, and kappa statistic. You must also show the weighted average of each measure (see an example format below).
      - Any other additional measures as needed.

    Do not include the screenshots of confusion matrices and performance measures from your program output in your report. Instead, you must create confusion matrices and performance measure tables in your report yourself.

    Example format of performance measure table:

	TPR	FPR	Precision	Recall	F-measure	ROC	MCC	Kappa
Class No								
Class Yes								
Wt. Average								

    - You must present your result using tables, graphs, charts, or in other visual format so that readers of your report can easily and effectively understand your result.
    - All parameters of your final best model (the best among 36 models).
    - Discussion and conclusion.
    - Division of work between team members.

- You must also submit the following datasets along with the final report:
    - The dataset after all your preprocessing. Name this *preprocessed_data.csv*.
    - The initial training and the test datasets. Name these files *initial_train.csv* and *initial_test.csv*, respectively.
    - If you created an intermediate file(s) that were used to build models, you must submit those file(s), if needed.

- Your final report must have at least 30 pages and at most 50 pages.
- Late submission penalty: 5 points will be deducted for each day.

<u>Presentation</u>

- The order of presentation will be determined randomly.
- All students must attend the presentation.
- If a student misses a presentation, 5 points will be deducted.

<u>Other Grading Criteria</u>

- If the final report is not well organized, up to 10 points will be deducted.
- If the description of all data mining procedure is not detailed enough, up to 10 points will be deducted.
- Your project results must be presented in your report effectively using tables and graphs so that readers of your report may understand the results easily and clearly. Otherwise, up to 10 points will be deducted.
- If your discussion/conclusion is not substantive up to 5 points will be deducted.
- If the final report does not include all required components, up to 10 points will be deducted.
- If all required datasets are not submitted, 5 points will be deducted.

Important:

- I should be able to reproduce your performance by running your code and following your description of data mining procedure (see the above criteria). So, it is very important that the description of your data mining procedure must be detailed and precise.
- You must submit a single R or Python code file. This code file must include all the steps you have taken for the project, including preprocessing. When I grade your project, I will run your code to try to reproduce your results.

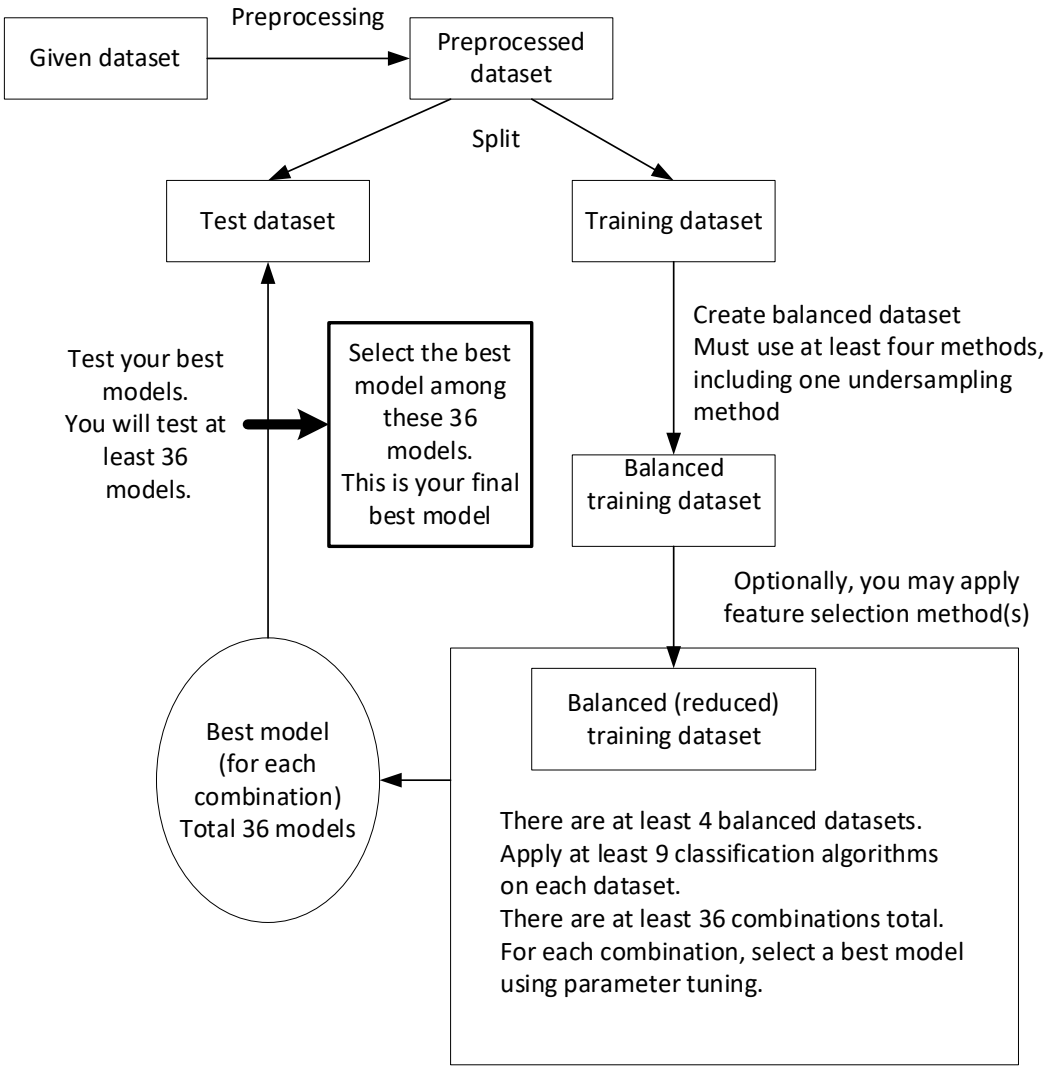
- In your code, you must include sufficient comments so that I can easily understand what each code segment does.
- **Additional requirement for writing a code is at the end of this assignment.**

<u>Deliverable and file naming convention</u>

- Final report: *LastName_FirstName_Report.pdf* or *LastName_FirstName_Report.docx*. If your team has two members, then include names of both members in the fine name.
- Dataset after preprocessing: *preprocessed_data.csv*
- Initial train and test datasets: *initial_train.csv* and *initial_test.csv*.
- Include all programs in a single file and name it *project_code.R*.

- When submitting the final report, combine the final report, datasets, R program file, and any other files you may have into ***a single archive file*** and name it *LastName_FirstName.<ext>*, where *<ext>* is an appropriate file extension, such as z*ip* or *rar*. Again, include names of both team members. Submit the archive file. **Do not submit individual files.**
- **Only one team member (not both) must submit all deliverables.**

Overall Project Process (simplified)



Note: Do noy use supervised dimension reduction methods during preprocessing (before split).

<u>Project code submission requirement (for both R and Python)</u>:

Grading projects takes a very long time. Often in the past it took me more than a half of a day to grade just one project. To be able to finish all grading on time I need your help.

- Bottom line: I should be able to **reproduce** results of your individual models separately so that I may be able to verify your performance for each model.

- Include all your programs in a single R or Python file. **Do not submit multiple code files.**
- If you use R, you must submit a *.R* file. **Do not submit a *.RMD* file**.
- If you use Python, you must submit a *.py* file. **Do not submit a notebook file (e.g., *.ipynb* file).**
- Your code must begin with reading *project_data.csv* file.
- Your code must include all preprocessing you performed.
- You are required to build at least 36 models with combinations of 4 balancing methods and 9 classification algorithms. **Do not build and test all 36 models in a single statement, such as a loop**. Write one code segment for each model so that I can reproduce your results of individual model separately. Suppose, for example, that b1, b2, b3, and b4 are 4 balancing methods and c1, c2, c3, c4, c5, c6, c7, c8, and c9 are 9 classification methods. You **should not** write your code in the following way:

```
for b in (b1, b2, b3, b4)
    for c in (c1, c2, c3, c4, c5, c6, c7, c8, c9)
        build and test a model with (b, c)
```

The above example uses a loop for the purpose of illustration. You should not use any similar construct.

Instead, you must write your code in the following way:

```
for (b1, c1):
    build and test a model
for (b1, c2):
    build and test a model

. . .

for (b2, c1):
    build and test a model
for (b2, c2):
    build and test a model

. . .
and so on
```

In other words, you need to build and test models one at a time.

For each model, your output on the screen must include the **confusion matrix** and other performance measures of the test result.